



Smoking Health Analysis Dashboard

"Data • Insights • Decisions • Better Healthcare"

Project :

Smoking is one of the leading causes of preventable diseases and deaths worldwide. It significantly affects vital organs such as the lungs, heart, kidneys, and blood vessels, increasing the risk of chronic illnesses including cardiovascular disease, lung cancer, high blood pressure, and respiratory disorders.

Healthcare organizations collect large amounts of patient data, including smoking status, age, gender, Body Mass Index (BMI), blood pressure, cholesterol levels, and organ health. However, this data is often stored across multiple systems, making it difficult for doctors and hospital administrators to identify health trends and high-risk patients quickly.

Traditional reports are time-consuming and provide limited insights into how smoking influences overall patient health. As a result, healthcare professionals may find it challenging to make timely, data-driven decisions for prevention and treatment.

To address this challenge, the Smoking Health Analysis Dashboard was developed using Power BI. The dashboard integrates patient information into a single interactive platform that allows users to analyze smoking patterns and their impact on health. It enables doctors, hospital management, and healthcare analysts to monitor patient conditions, identify high-risk groups, compare health indicators across different populations, and support preventive healthcare strategies.

Problem Statement:

Hospitals and healthcare providers manage thousands of patient records every day. Analyzing the relationship between smoking habits and health conditions manually is difficult, time-consuming, and prone to errors.

Data Cleaning and Transformation:

Before creating the dashboard, the raw healthcare dataset was cleaned and transformed using Power Query Editor in Power BI to ensure accurate and reliable analysis.

Data Cleaning Steps

The following data cleaning activities were performed:

1. Imported the Dataset

- Loaded the healthcare dataset into Power BI from the source file (Excel/CSV).

2. Checked Data Types

- Verified that each column had the correct data type.
- Example:
 - Age → Whole Number
 - BMI → Decimal Number
 - Blood Pressure → Whole Number
 - Date → Date
 - Smoking Status → Text

3. Removed Duplicate Records

- Identified and removed duplicate patient records to avoid incorrect analysis.

4. Handled Missing Values

- Checked for blank or null values.
- Replaced missing values where appropriate or removed incomplete records.

5. Standardized Text Values

- Corrected inconsistent spellings and formatting.
- Examples:
 - Male, male, MALE → Male
 - Smoker, smoker → Smoker

6. Removed Unnecessary Columns

- Deleted columns that were not required for dashboard analysis.

7. Renamed Columns

- Renamed column headers to meaningful names for easier understanding.

8. Created New Columns (if required)

Examples:

- BMI Category
- Age Group
- Blood Pressure Category
- Cholesterol Category

9. Verified Data Quality

- Ensured the dataset was clean, consistent, and ready for analysis.

DAX Measures Created

- Several DAX measures were created to calculate KPIs and summarize healthcare information.
- **Total Patients**
 - `Total Patients = COUNT(Patients[Patient ID])`
 - **Purpose:**
Calculates the total number of patients.
- **Average BMI**
 - `Average BMI = AVERAGE(Patients[BMI])`
 - **Purpose:**
Calculates the average BMI of all patients.
- **Average Blood Pressure**
 - `Average BP = AVERAGE(Patients[Blood Pressure])`
 - **Purpose:**
Calculates the average blood pressure.
- **Average Cholesterol**
 - `Average Cholesterol = AVERAGE(Patients[Cholesterol])`
 - **Purpose:**
Shows the average cholesterol level.
- **Total Smokers**
 - `Total Smokers =`
 - `CALCULATE (`
 - `COUNT(Patients[Patient ID]),`
 - `Patients[Smoking Status]="Smoker"`
 - `)`
 - **Purpose:**
Counts the total number of smokers.
- **Smoking Percentage**
 - `Smoking % =`
 - `DIVIDE (`
 - `[Total Smokers],`
 - `[Total Patients],`
 - `0`

-)
- Purpose:
Calculates the percentage of smokers.
- Average Age
- Average Age = AVERAGE(Patients[Age])
- Purpose:
Shows the average age of patients.

DAX Measures Summary

- To support data analysis and create dynamic Key Performance Indicators (KPIs), several DAX measures were developed in Power BI. These measures automatically calculate important healthcare metrics and update based on user-selected filters and slicers. The measures include **Total Patients**, which counts the total number of patient records; **Average Age**, which calculates the average age of patients; **Average BMI**, which measures the average Body Mass Index; **Average Blood Pressure**, which calculates the average blood pressure level; **Average Cholesterol**, which determines the average cholesterol level; **Total Smokers**, which counts all patients identified as smokers; and **Smoking Percentage**, which calculates the proportion of smokers among the total patient population. Together, these DAX measures provide accurate, real-time insights into patient demographics, health conditions, and smoking-related statistics, enabling healthcare professionals to monitor performance, identify high-risk groups, and make informed decisions through interactive Power BI dashboards.

Key Insights

1. Smoking is more common in specific age groups, indicating that targeted awareness and prevention programs should focus on these populations.
2. Patients who smoke generally have higher BMI, blood pressure, and cholesterol levels, suggesting a greater risk of developing chronic diseases such as heart disease and stroke.
3. Organ health is significantly affected by smoking, with smokers showing a higher proportion of damaged organs compared to non-smokers.
4. Male and female smoking patterns differ, highlighting the need for gender-specific health education and intervention strategies.
5. High-risk patients can be identified by combining smoking status with BMI, blood pressure, and cholesterol levels, enabling healthcare providers to prioritize preventive care.
6. Interactive filtering allows healthcare professionals to analyze patient groups by age, gender, smoking status, and organ condition, making it easier to identify trends and support clinical decision-making.

Recommendations:

- 1) **Strengthen Smoking Cessation Programs:** Identify patients who are current smokers and provide counseling, nicotine replacement therapy, and regular follow-up programs to help reduce smoking-related health risks.
- 2) **Prioritize High-Risk Patients:** Use the dashboard to identify patients with high BMI, elevated blood pressure, high cholesterol, and smoking habits. These patients should receive regular health screenings and personalized treatment plans.
- 3) **Promote Preventive Healthcare:** Conduct awareness campaigns encouraging healthy lifestyles, balanced nutrition, regular physical activity, and smoking cessation to reduce the incidence of chronic diseases.
- 4) **Implement Routine Health Monitoring:** Schedule periodic health checkups for smokers and high-risk individuals to monitor changes in vital health indicators and detect diseases at an early stage.
- 5) **Support Data-Driven Decision Making:** Hospital administrators should use the dashboard to monitor patient trends, allocate medical resources efficiently, and evaluate the effectiveness of healthcare programs.
- 6) **Enhance Patient Education:** Educate patients about the long-term impact of smoking on organs such as the lungs, heart, and kidneys, helping them make informed lifestyle choices.
- 7) **Improve Resource Allocation:** Analyze department-wise patient trends to ensure adequate staffing, medical equipment, and healthcare resources are available where demand is highest.
- 8) **Develop Personalized Care Plans:** Combine smoking status with BMI, cholesterol, blood pressure, and age to create individualized treatment and preventive care strategies for each patient.
- 9) **Monitor Healthcare Performance:** Regularly review dashboard KPIs to measure improvements in patient health, evaluate smoking reduction initiatives, and assess overall hospital performance.
- 10) **Leverage Predictive Analytics:** In future versions, integrate machine learning models with the dashboard to predict disease risk, identify vulnerable patient groups, and enable proactive healthcare interventions.

Overall Insight

The **Smoking Health Analysis Dashboard** transforms raw patient data into meaningful healthcare insights. It enables hospitals to identify smoking-related health risks, monitor patient conditions, support preventive healthcare initiatives, and make informed decisions that improve patient outcomes and operational efficiency.

Conclusion:

Implementing these recommendations will help healthcare organizations reduce smoking-related health complications, improve patient outcomes, optimize hospital operations, and support evidence-based clinical decision-making. The Power BI dashboard serves as a valuable decision-support tool by transforming healthcare data into meaningful insights that enable timely and effective actions.

**"Transforming Healthcare
Data into Smarter Decisions, Healthier Brain for good thoughts".**



THANK YOU MAM.